



Build with Vertex AI Skill Badge Sessions

Short virtual training to introduce learners to the technical skill badge credential on building with Vertex AI and provide an overview of its respective challenge labs.

 **DURATION**
3 hours per badge

 **LEVEL**
Advanced

 **FORMAT**
Instructor led

What you'll learn

- Leverage extended Gemini API capabilities.
- Develop solutions using Model Garden models and APIs.
- Understand, generate, and customize media content with Vertex AI.
- Build RAG applications with Vertex AI.
- Evaluate model and agent performance.



Overview	5 modules · 5 labs
Who this course is for	Technical individuals looking to enhance their understanding of gen AI, stay updated with technological advancements, and validate their expertise.
Products	Vertex AI, Gemini, Agent Development Kit
Prerequisite	Learners should be familiar with Google Cloud. It is recommended learners begin with the first skill badge in this series.
Not covered	Theory and respective learning path content.

Module 01 Extend Gemini with controlled generation and tool use

Topics	Complete the “Extend Gemini with controlled generation and tool use” skill badge to demonstrate your proficiency in connecting models to external tools and APIs. This allows models to augment their knowledge, extend their capabilities, and interact with external systems to take actions such as sending an email.
Objectives	Demonstrate your ability to implement the ReAct pattern by extending LLMs to access external information, such as databases, third party services, and a coding environment.
Activities	Slideware and a challenge lab

Module 02 Build gen AI solutions using Model Garden models and APIs

Topics	Complete the “Build gen AI solutions using Model Garden models and APIs” skill badge to demonstrate your ability to use Vertex AI Model Garden features when building gen AI solutions. You will use partner APIs, such as Anthropic Claude and Meta Llama, deploy and programmatically access foundation models, like Gemma and Stable Diffusion XL, and access Vertex AI Endpoints.
Objectives	Demonstrate your ability to use Vertex AI Model Garden features when building gen AI solutions.
Activities	Slideware and a challenge lab



Module 03 Edit images with Imagen

Topics	Complete the “Edit images with Imagen” skill badge to demonstrate your skills with Imagen's mask modes and editing modes to edit images according to certain prompts.
Objectives	• Demonstrate your ability to edit and enhance images with generative AI without distorting the product featured in the image. • Be able to implement outpainting, mask-free editing, inpainting insertion, inpainting removal, and background swapping.
Activities	Slideware and a challenge lab

Module 04 Deploy a RAG application with vector search in Firestore

Topics	Complete this skill badge to show your ability to develop a real-world generative AI Q&A solution using a RAG framework. You will use Firestore as a vector database and deploy a Flask app as a user interface to query a food safety knowledge base.
Objectives	• Be able to develop a real-world generative AI Q&A solution using a RAG framework. • Use Firestore as a vector database. • Deploy a Flask app as a user interface to query a food safety knowledge base. • Demonstrate your ability to load a text document and split it into chunks, generate embeddings for each chunk, store the text chunks and their embeddings, conduct vector search to return similar documents to a query document.
Activities	Slideware and a challenge lab

Module 05 Evaluate gen AI model and agent performance

Topics	Complete the “Evaluate gen AI model and agent performance” skill badge to demonstrate your ability to use the gen AI evaluation service. You will evaluate models to select the best model for a given task, compare models against each other, and evaluate the performance of agents.
Objectives	• Test your ability to evaluate generative model and agent performance. • Be able to use the Gen AI Evaluation Service to complete this evaluation.
Activities	Slideware and a challenge lab

